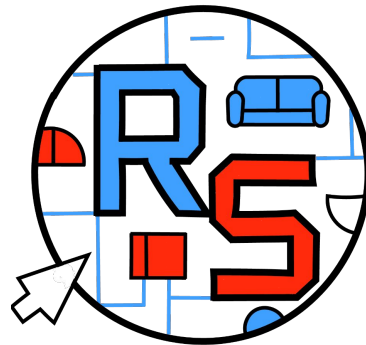


RoomStudio



Randy Nguyen, Cody Nutter, Gavin Bonham

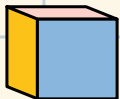
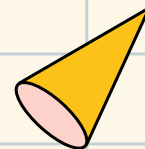
Advisor: Dr. Wolff

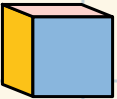


Table of Contents

x

- Our Idea
- Requirements & Our Process
- Development & Tech Overview
- Demo of our app and features
- Future Goals





The Problem



Picking Out Furniture Is:

- Time Consuming
- Expensive
- Labor Intensive

Designing Ideas Is Hard!

- How do you know what will fit in your room?
- How do you know what options you have for furniture?

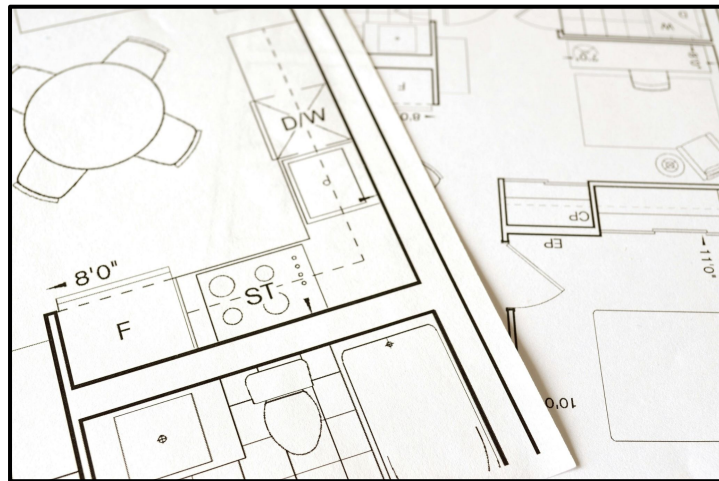


Our Idea

An interactive desktop application for customizing and editing floor plans.

Offers:

- *Easy visualization*
- *Real-world accurate measurements*
- *Rapid prototyping*
- *Furniture lookup*





User Stories

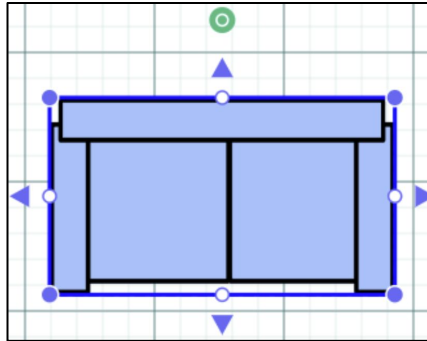
- **As a homeowner, I want a way to easily visualize how I can fill my space.**
 - *2D visuals that you can use to see plan available space in a room*
 - *Allow you to move furniture items around the virtual space*
 - *Visualize a room plan to see your space with accurate dimensions of furniture items*
- **As an interior designer, I want to efficiently search for different furniture that fits a specific need.**
 - *Have a search bar for searching specific furniture items from the Amazon catalog*
 - *Be able to view images of furniture and specs like size*

Requirements / Goals

2D Canvas

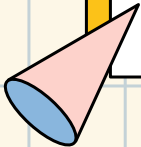
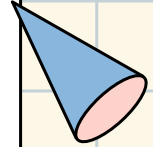


**Add/Remove/
Modify Furniture**



**Save/Load +
Furniture Search**

amazon





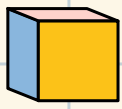
Design Decisions and Pivoting

- **Initial Design Ideas**

- *Web application with database for users and room designs*
- *Allow for 3D rendering with 3D models to see your space*
- *Having AI you can prompt to organize your room for you*

- ***Pivoting***

- *Decided to create it as desktop application*
- *Focused on 2D as visualizaiton*
- *Moved away from AI tools, out of scope for current time period,*

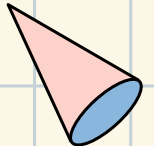


Development Process



- Used Git for version control
- Created a list of priorities for feature implementation
- Each member given different feature to work on
- Once a week check in on progress and update goals
- Communicated Using Discord

Priorities 		
Pos. # 	☒ Done? 	Features (ranked top to bottom) 
1	☒	Select Objects
2	☒	Moving Objects
3	☒	Add objects
4	☐	Rotating Objects
5	☐	Measuring Stick
6	☐	Measuring Scale
7	☐	Scaling Objects
8	☐	Draw room shape(s)





Development Timeline

- **Initial Draft**
 - *Deciding on scope and development tools and drafting diagrams and UI ideas*
- **Canvas Implementation**
 - *Testing different implementation for canvas and working on object drawing implementation*
- **Important Interactivity**
 - *Worked in critical features like the ability to add, select, and edit furniture items, along with Implementation of panning and zooming*
- **More refined and critical features**
 - *Walls, loading room layouts*
- **Polish and more QOL features**
 - *Doors and windows, saving to PDF*
 - *More hotkeys, easier furniture manipulation*



Tech Stack

x

Electron

- Cross Platform
- Flexible

TypeScript

- Static Typing
- Enhanced tooling

React

- Component-based state mgmt.

HTML/CSS

- Used for canvas and UI design

NodeJS

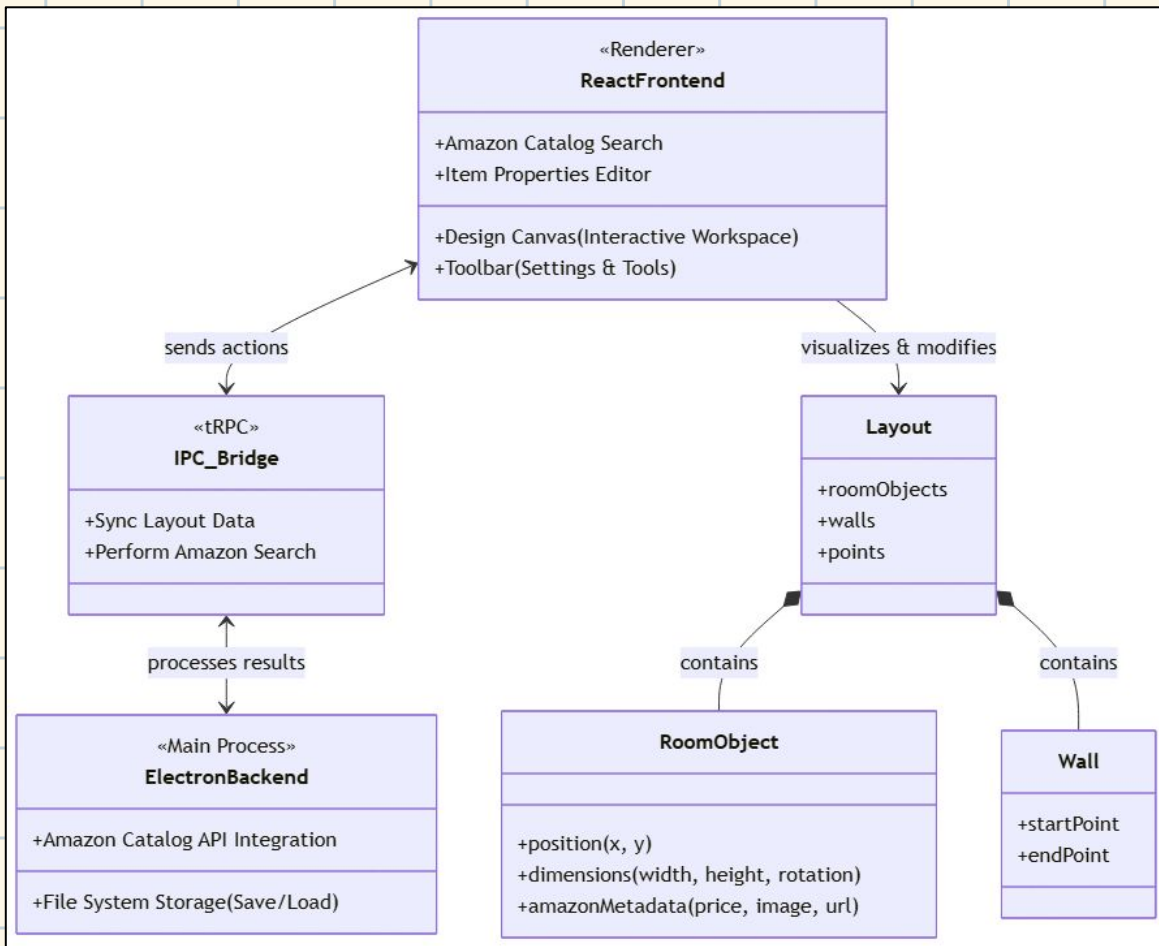
- API Management
- Versatile

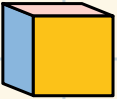
Amazon SP-API

- Sellers catalog
- Dynamic data



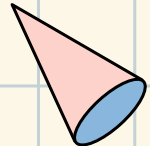
Diagram of Application





Testing

- **Worked closely and demoed the app with Design/Art students**
 - *Workflow of RoomStudio similar to PhotoShop, and other design/art apps*
 - *Got feedback on feel of the application and the tools available*

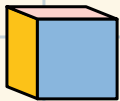
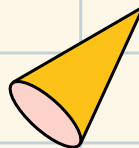




Challenges

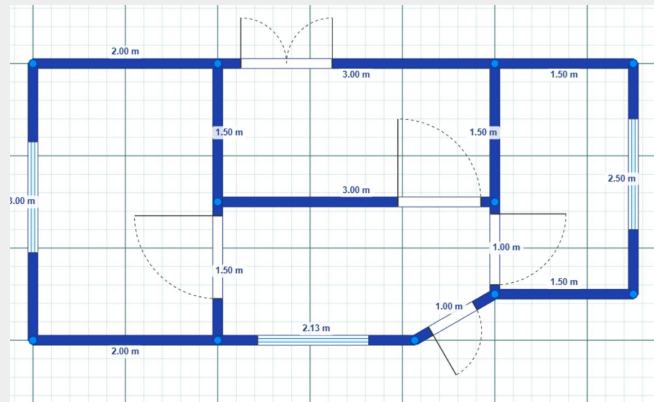
x

- Deciding on setup for application and getting the framework setup took more time than expected
- Deciding how to implement actual canvas
- Some feature implementations took longer than expected
- Figuring out how to implement things with maintainability in mind



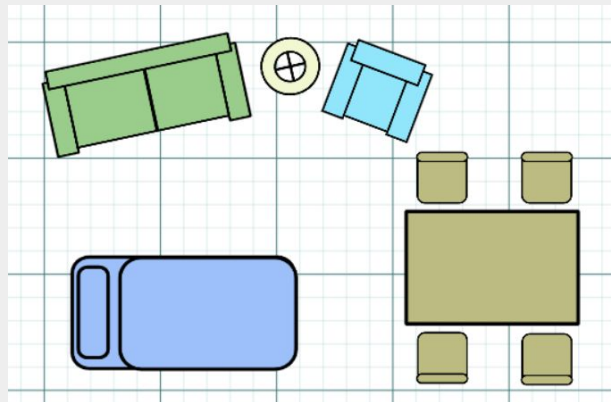
Design Systems

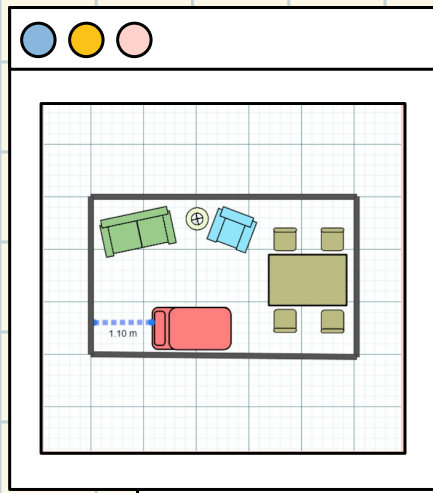
Wall Layout



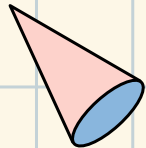
+

Furniture Layout





Demo Time!





Future Goals

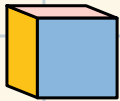
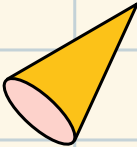
x

- **More tools and features**

- *More complex furniture types and more unique images*
- *More freedom with wall placement*
- *Leaning more into Product Sourcing aspects*

- **More User testing**

- *Finding ways to make more user friendly, keybinds*
- *See what other features might be useful*





AI Usage

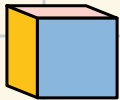
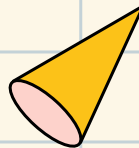
x

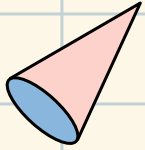
What we used AI for:

- Looking up documentation
- Debugging

What we did NOT use AI for:

- Writing code without understanding
- User Interface





Thank You!

x

Special Thanks to:
Our advisor Advisor: Dr. Wolff
Design students Chris and Gabby

